

## NICU PHARMACOLOGY

# CAFFEINE vs AMINOPHYLLINE

## Why Is Caffeine Preferred for Apnea of Prematurity?

A practical, evidence-informed comparison for neonatal nursing and clinical education

**KEY CLINICAL MESSAGE**

Caffeine citrate is the preferred methylxanthine for apnea of prematurity (AOP). Aminophylline/theophylline may be used as an alternative when caffeine is unavailable or when local protocols specifically indicate it.

## 1. Key Clinical Message

- Caffeine citrate is the preferred methylxanthine for apnea of prematurity (AOP).
- Aminophylline/theophylline may be used as an alternative when caffeine is unavailable or when local protocols specifically indicate it.
- Caffeine is easier to dose, has a wider therapeutic margin, and generally requires less routine serum-level monitoring.
- Its long half-life supports once-daily dosing and makes treatment simpler in the NICU and, when appropriate, after discharge.
- **Clinical practice should follow the neonatal unit protocol, formulary, and prescriber guidance.**

## 2. Why Caffeine Is Preferred

| Feature            | Caffeine citrate                       | Aminophylline / theophylline                                         |
|--------------------|----------------------------------------|----------------------------------------------------------------------|
| Role in AOP        | Preferred / first-line methylxanthine  | Alternative methylxanthine                                           |
| Therapeutic margin | Wider; lower toxicity concern          | Narrower; toxicity is more concerning                                |
| Half-life          | Long in preterm infants                | Shorter; more frequent dosing                                        |
| Dosing convenience | Usually once daily after loading       | Usually requires more frequent dosing                                |
| Serum levels       | Not routinely required in most infants | More often considered when toxicity or unusual response is suspected |
| Evidence base      | Extensively studied in AOP             | Less favorable evidence base for routine AOP use                     |

## 3. Efficacy and Clinical Outcomes

- Reduces the frequency of apnea and intermittent hypoxemia in preterm infants.
- Facilitates successful extubation and reduces the need for respiratory support in appropriate infants.
- Caffeine treatment has been associated with reduced bronchopulmonary dysplasia (BPD) in very-low-birth-weight infants.
- The Caffeine for Apnea of Prematurity (CAP) trial and subsequent follow-up studies provide strong evidence supporting caffeine therapy.
- Later neurodevelopmental outcomes are more nuanced: early benefits were reported, but not every advantage persisted at longer follow-up.

**Bottom line:** Caffeine has the strongest overall evidence and the best practical profile for routine AOP management.

## 4. Safety and Monitoring

- Caffeine has a wider therapeutic window than aminophylline/theophylline.
- Common adverse effects may include tachycardia, feeding intolerance, restlessness/jitteriness, and increased metabolic activity.
- Routine serum caffeine levels are generally unnecessary in stable infants receiving standard doses.
- Monitor clinically: heart rate, respiratory pattern/apnea, oxygen requirement, feeding tolerance, and signs of toxicity.
- Consider serum drug levels when unusually high doses are used, toxicity is suspected, renal/hepatic issues affect clearance, or the clinical response is unexpected.
- Therapeutic serum ranges vary by reference and clinical context; when levels are indicated, follow the local laboratory and NICU protocol.

**Clinical pearl:** “Less monitoring” does not mean “no monitoring.” Neonates remain excellent at turning small dosing errors into large problems.

## 5. Aminophylline: Why It Is Generally Not First-Line

- Aminophylline is effective against apnea because it contains theophylline, a methylxanthine respiratory stimulant.
- Its narrower therapeutic index makes dosing and toxicity management less forgiving.
- Adverse effects can include tachycardia, feeding intolerance, vomiting, irritability, and other signs of methylxanthine toxicity.
- Drug-level monitoring may be needed more often, depending on the clinical situation and local protocol.
- Older neonatal literature raised concerns about effects on cerebral blood flow and gastrointestinal tolerance. These concerns contribute to its less favorable profile compared with caffeine.

**Clinical Pearl:** BPD alone does not establish an indication for aminophylline or bronchodilator therapy. Treatment should be guided by the infant’s clinical condition and local protocol.

## 6. Practical NICU Comparison

| Parameter         | Caffeine            | Aminophylline                   | NICU implication                            |
|-------------------|---------------------|---------------------------------|---------------------------------------------|
| Typical AOP role  | Preferred           | Alternative                     | Use caffeine when available and appropriate |
| Dosing frequency  | Usually once daily  | More frequent                   | Caffeine simplifies medication rounds       |
| Therapeutic index | Wider               | Narrower                        | Lower risk of dosing-related toxicity       |
| Monitoring        | Mostly clinical     | Clinical + level when indicated | Follow unit protocol                        |
| Long half-life    | Yes                 | No (shorter half-life)          | Supports sustained effect                   |
| Evidence          | Strong AOP evidence | Older/less robust evidence      | Caffeine is generally favored               |

## 7. Bedside Nursing Focus

- **Medication safety:** Verify the prescribed caffeine preparation and dose carefully.
- **Formulation check:** Caffeine citrate and caffeine base are not numerically interchangeable without accounting for the formulation. Caffeine citrate contains 50% caffeine base; 20 mg caffeine citrate is equivalent to 10 mg caffeine base.
- **Medication reconciliation:** Confirm whether the prescribed dose is expressed as caffeine citrate or caffeine base before administration.
- **Documentation:** Record loading and maintenance doses clearly to prevent duplicate administration.
- **Event tracking:** Monitor apnea/bradycardia/desaturation events and document frequency, duration, and interventions.
- **Clinical trending:** Trend heart rate and oxygen requirement after administration.
- **GI assessment:** Assess feeding tolerance, abdominal findings, emesis, and stool pattern as part of routine neonatal assessment.
- **Escalation:** Escalate persistent apnea despite therapy, significant bradycardia/desaturation, or suspected toxicity promptly.

**Always use the NICU's current medication protocol for exact dose, dilution, route, and administration timing.**

## 8. Take-Home Message

### CAFFEINE: PRACTICALITY + SAFETY + EVIDENCE

- Caffeine is generally preferred for apnea of prematurity.
- Its long half-life supports convenient once-daily dosing.
- Its wider therapeutic margin provides a more forgiving dosing profile than aminophylline/theophylline.
- Strong evidence supports its effectiveness in reducing apnea and improving important respiratory outcomes.
- Aminophylline remains an alternative when caffeine cannot be used or is unavailable, according to local practice.

Clinical decisions must follow current neonatal guidelines, local protocols, and the responsible clinician's prescription.

## 9. Selected References

1. Medscape. Apnea of Prematurity: treatment and medication overview.
2. Medscape. Apnea of Prematurity: Medication.
3. Schmidt B, et al. Caffeine Therapy for Apnea of Prematurity. New England Journal of Medicine. 2006;354:2113–2121.
4. Cochrane / systematic review literature on caffeine for apnea of prematurity and bronchopulmonary dysplasia.

5. Recent meta-analysis: PubMed PMID 38553606 (2024).
6. High-dose caffeine safety literature: PubMed PMID 34290376.

**Educational material only.** Medication decisions must follow current neonatal guidelines, local protocols, and the responsible clinician's prescription.